Cycloidal Curves Formed by Rolling a Circle on the Surface of a Sphere

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Abstract

This paper introduces general formulae for cycloidal curves formed by rolling a circle along a given path on the surface of a sphere.

1 Preface

I originally wrote this paper sometime around late 2013/early 2014 during my undergrad and attempted to have it published, but the journal to which I submitted it rejected it for reasons I don't fully recall. While I have gone through and checked for errors, this paper has not been peer-reviewed. If you notice any errors, please let me know!

2 Introduction

A cycloid is the path traced out by a point on the edge of a circle as it rolls in straight line [1]. Similarly, epicycloids, hypocycloids, and other such curves describe the path traced out by a point on a circle as it rolls along the edge of some other circle [2][3][4][5]. Considering the popularity of cycloidal curves, it seems only natural that an equation should exist to describe the path traced out by a point on a circle as that circle rolls along some given path on the surface of a sphere. However, any amount of time spent searching for such an equation reveals that if these formulae do exist, they are indeed obscure. This paper derives a general formula describing such spherical epicycloids, then presents formulae for finding spherical hypocycloids, spherical epitrochoids, and spherical hypotrochoids.

3 Deriving the General Formula for a Spherical Epicycloid

We wish to describe the path traced out by a point on a circle as it rolls on, and adjacent to, the surface of a sphere. We begin by choosing an equation that describes the path traced out by the center of the circle as it travels on the sphere. Since the radius of the sphere and circle are constant, we choose to begin with a set of equations describing a sphere of radius $R + r$:

$$\begin{aligned} x &= (R + r)\cos(\theta)\cos(\phi) \\ y &= (R + r)\sin(\theta)\cos(\phi) \\ z &= (R + r)\sin(\phi) \end{aligned} \tag{1}$$

where, R is the radius of our sphere, r is the radius of our circle, θ (a function of t) tells us how far we have rotated in the xy plane, and ϕ (also a function of t) tells us the angle we have rotated off the xy plane. This function describes the path that the center of our circle takes as it travels around our sphere. We can rewrite the equation as a vector like so:

$$r(t) = (R+r)\cos(\theta)\cos(\phi)\hat{\imath} + (R+r)\sin(\theta)\cos(\phi)\hat{\jmath} + (R+r)\sin(\phi)\hat{k} \tag{2}$$

In order to know which way to orient the circle as it rolls around the sphere, we need to find two particular vectors. One of these vectors must always point directly away from the center of the sphere, and must have a length of 1. We'll call this vector $\mathbf{N}$. The other vector we need to find is the unit tangent vector $\mathbf{T}$ for the vector $r(t)$. By definition, $\mathbf{T}$ will always point in the direction in which the center of our circle is traveling, and will always have a magnitude of 1. To find $\mathbf{T}$, we begin by taking the derivative of the vector $r(t)$. Thus we find:

$$\begin{gathered} v(t) = \\ -(R+r)\left(\frac{\partial\theta}{\partial t}\right)\sin(\theta)\cos(\phi) - (R+r)\left(\frac{\partial\phi}{\partial t}\right)\cos(\theta)\sin(\phi)\hat{\imath} \\ +(R+r)\left(\frac{\partial\theta}{\partial t}\right)\cos(\theta)\cos(\phi) - (R+r)\left(\frac{\partial\phi}{\partial t}\right)\sin(\theta)\sin(\phi)\hat{\jmath} \\ +(R+r)\left(\frac{\partial\phi}{\partial t}\right)\cos(\phi)\hat{k} \end{gathered} \tag{3}$$

Next we need to find the magnitude of this vector, which simplifies nicely to:

$$\|v(t)\| = (R+r)\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2\cos^2(\phi)} \tag{4}$$

Thus we find $\mathbf{T}$ to be:

$$\begin{gathered} \mathbf{T} = \\ -\frac{\left(\frac{\partial\theta}{\partial t}\right)\sin(\theta)\cos(\phi) + \left(\frac{\partial\phi}{\partial t}\right)\cos(\theta)\sin(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2\cos^2(\phi)}}\hat{\imath} \\ +\frac{\left(\frac{\partial\theta}{\partial t}\right)\cos(\theta)\cos(\phi) - \left(\frac{\partial\phi}{\partial t}\right)\sin(\theta)\sin(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2\cos^2(\phi)}}\hat{\jmath} \\ +\frac{\left(\frac{\partial\phi}{\partial t}\right)\cos(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2\cos^2(\phi)}}\hat{k} \end{gathered} \tag{5}$$

Since $\mathbf{N}$ is essentially the unit normal vector for a sphere of radius $R+r$, $\mathbf{N}$ is simply:

$$\mathbf{N} = \cos(\theta)\cos(\phi)\hat{\imath} + \sin(\theta)\cos(\phi)\hat{\jmath} + \sin(\phi)\hat{k} \tag{6}$$

A very important property of $\mathbf{N}$ and $\mathbf{T}$ is that they are always orthogonal to each other. Imagine a standard Cartesian plane with two vectors of length 1 situated at the origin with one vector pointing directly in the positive x direction and the other pointing directly in the positive y direction. If we wanted to obtain a vector starting at the origin of this Cartesian plane and going to some point (a, b) also in the same Cartesian plane, we could simply multiply our x vector by a and our y vector by b then add the two vectors together. Similarly, if we choose to position $\mathbf{N}$ and $\mathbf{T}$ so that they both start at the end of our vector $r(t)$, we can use them as a sort-of moving Cartesian plane, with $\mathbf{N}$ serving as our x-axis and $\mathbf{T}$ serving as our y-axis.

Since we wish to plot a circle onto this new "Cartesian plane" we should consider the following parametric equation for a circle of radius r:

$$\begin{aligned} x &= r\cos(\alpha) \\ y &= r\sin(\alpha) \end{aligned} \tag{7}$$

Where α is the angle our circle has rotated in our moving Cartesian plane at time t. As a matter of preference, we can choose to use a slightly modified version of the above equation:

$$\begin{aligned} x &= -r\cos(\alpha) \\ y &= -r\sin(\alpha) \end{aligned} \tag{8}$$

Changing both of these equations to be negative simply changes our starting position on the circle. While both of these equations will work when deriving our equation, we will be using Equation 8 for the remainder of the paper.

In order to plot the above parametric equation into our moving Cartesian plane, we need to multiply x and y by $\mathbf{N}$ and $\mathbf{T}$ respectively. This gives us:

$$\begin{gathered} \mathbf{N}_x = \\ -r\cos(\alpha)\cos(\theta)\cos(\phi)\,\hat{\imath} \\ -r\cos(\alpha)\sin(\theta)\cos(\phi)\,\hat{\jmath} \\ -r\cos(\alpha)\sin(\phi)\,\hat{k} \end{gathered} \tag{9}$$

And:

$$\begin{gathered} \mathbf{T}_y = \\ r\sin(\alpha)\frac{\left(\frac{\partial\theta}{\partial t}\right)\sin(\theta)\cos(\phi) + \left(\frac{\partial\phi}{\partial t}\right)\cos(\theta)\sin(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2\cos^2(\phi)}}\,\hat{\imath} \\ -r\sin(\alpha)\frac{\left(\frac{\partial\theta}{\partial t}\right)\cos(\theta)\cos(\phi) - \left(\frac{\partial\phi}{\partial t}\right)\sin(\theta)\sin(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2\cos^2(\phi)}}\,\hat{\jmath} \\ -r\sin(\alpha)\frac{\left(\frac{\partial\phi}{\partial t}\right)\cos(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2\cos^2(\phi)}}\,\hat{k} \end{gathered} \tag{10}$$

At this point we need to find a way to calculate α. To help visualize the next portion of the derivation, imagine we have a coin that has recently been rolled on an inkpad so that its edge is covered in a still wet coat of ink. If we were to take the freshly inked coin and roll it on a piece of paper, the length of the line left by the ink on the piece of paper would be equal to the radius of the coin times the angle the coin had rotated. Now imagine we were to coat the outer edge of our circle in ink then roll it on the surface of our sphere. The length of the line, L, left on the surface of the sphere would be equal to r, the radius of our circle, times α, the angle our circle had rotated in our moving Cartesian plane. Mathematically, this is represented by:

$$L = r\alpha \tag{11}$$

Thus:

$$\alpha = \frac{L}{r} \tag{12}$$

We can find L by finding the arc length of the following curve at time t:

$$\begin{aligned} x &= R\cos(\theta)\cos(\phi) \\ y &= R\sin(\theta)\cos(\phi) \\ z &= R\sin(\phi) \end{aligned} \tag{13}$$

To find this length, we introduce θ' and ϕ'. These functions are equal to θ and ϕ respectively, where t has been substituted by the dummy variable, τ. Using these new equations, we can find the arc length on the surface of the sphere at time t with the following equation:

$$L = R\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau \tag{14}$$

Where a is the starting value for t. Substituting this into Equation 12 gives us:

$$\alpha = \frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau \tag{15}$$

All that's left to do now is to substitute α in $\mathbf{N}_x$ and $\mathbf{T}_y$ with the above equation, then add the vectors $\mathbf{N}_x$ and $\mathbf{T}_y$ to our original vector $r(t)$. This new equation, which we shall label S_c is shown on the following page with some minor simplification.

$$
\begin{aligned}
S_c = \\
&\cos(\theta)\cos(\phi)\left((R+r) - r\cos\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\right) \\
&+r\sin\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\frac{\left(\frac{\partial\theta}{\partial t}\right)\sin(\theta)\cos(\phi) + \left(\frac{\partial\phi}{\partial t}\right)\cos(\theta)\sin(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2 \cos^2(\phi)}}\,\hat{\imath} \\
&+\sin(\theta)\cos(\phi)\left((R+r) - r\cos\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\right) \\
&-r\sin\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\frac{\left(\frac{\partial\theta}{\partial t}\right)\cos(\theta)\cos(\phi) - \left(\frac{\partial\phi}{\partial t}\right)\sin(\theta)\sin(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2 \cos^2(\phi)}}\,\hat{\jmath} \\
&+\sin(\phi)\left((R+r) - r\cos\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\right) \\
&-r\sin\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\frac{\left(\frac{\partial\phi}{\partial t}\right)\cos(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2 \cos^2(\phi)}}\,\hat{k}
\end{aligned}
\tag{16}
$$

where $a \leq t \leq b$

4 Examples of Spherical Epicycloids

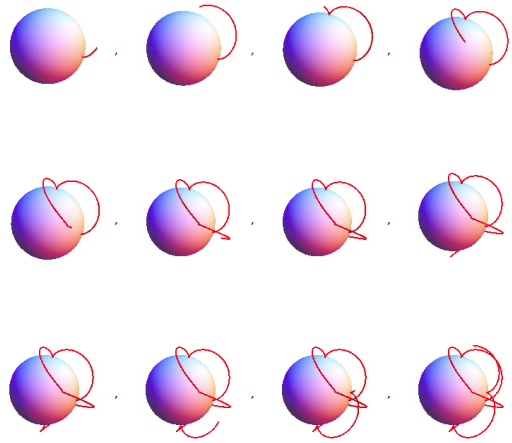


Figure 1: This series of images shows the curve formed by our equation when $R = 4$, $r = 1$, and $\theta = \phi = t$. This image demonstrates the progression of the curve from $t = 0$ to $t = \frac{23\pi}{10}$. The sphere is not part of the graph, and has only been included as a visual aide.

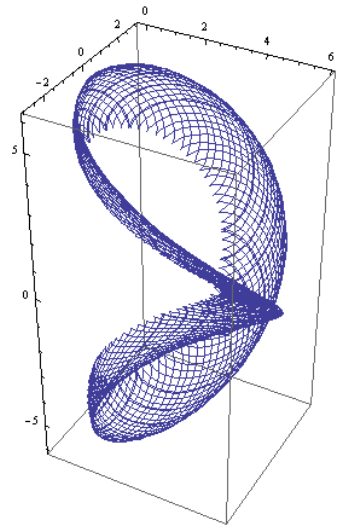


Figure 2: This image shows the graph formed using the same parameters as those used in Figure 1, but with $0 \leq t \leq 50\pi$.

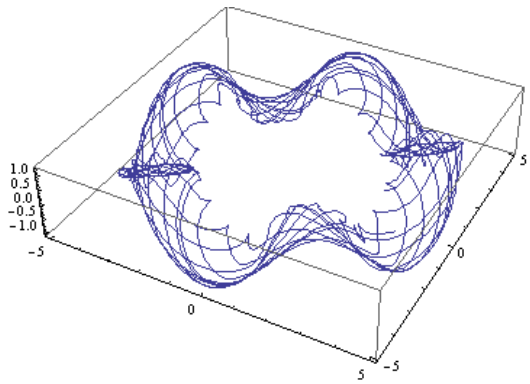


Figure 3: This graph was formed using the parameters $R = 3$, $r = 1$, $\theta = t$, and $\phi = \frac{1}{5}\sin(5t)$ with $0 \leq t \leq 20\pi$.

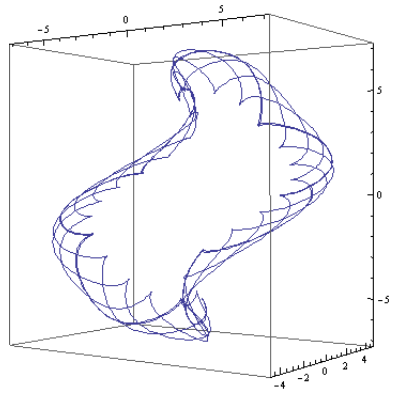


Figure 4: The graph formed when $R = 5$, $r = 1$, $\theta = \sin(t)$, and $\phi = \frac{t}{2}$ with $0 \leq t \leq 20\pi$.

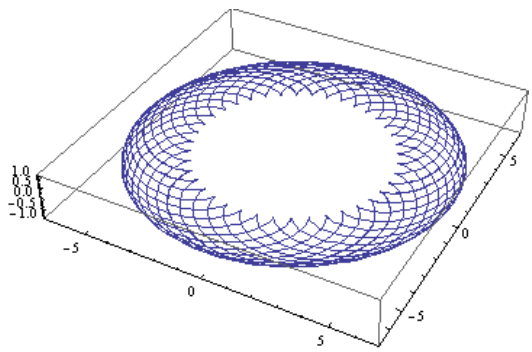


Figure 5: As one might expect, setting ϕ equal to zero allows us to create regular epicycloids on the xy plane. This graph was created using the parameters $R = 4$, $r = 1.3$, $\theta = t$, and $\phi = 0$ with $0 \leq t \leq 30\pi$.

5 Examples of Spherical Hypocycloids

A hypocycloid is formed in a similar manner to an epicycloid but with a circle rolling on the inside (contrary to outside) of another circle. If we choose a value for r such that r is negative, our equation will create a spherical hypocycloid. Alternatively, we could alter the equation S_c such that all instances of r are negative. Two examples of spherical hypocycloids are presented below.

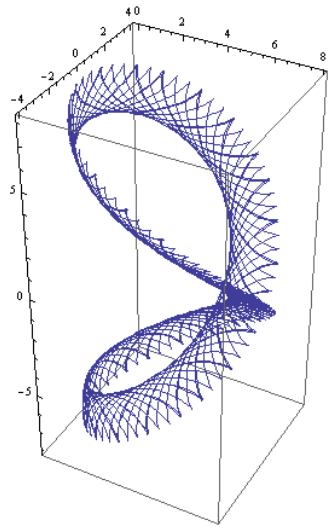


Figure 6: The graph formed by our equation, S_c, when $R = 8$, $r = -1$, and $\theta = \phi = t$ when $0 \leq t \leq 30\pi$.

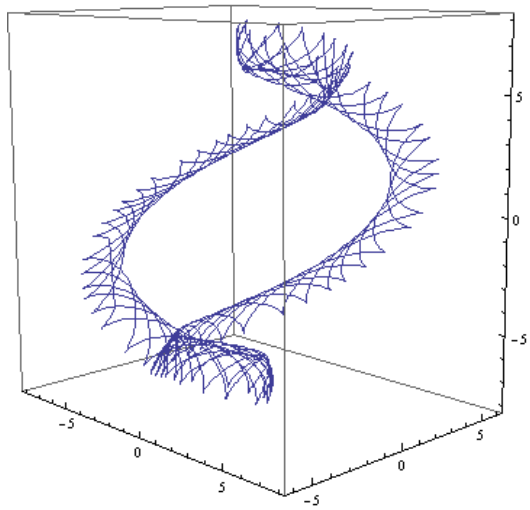


Figure 7: The graph formed by our equation, S_c, when $R = 8$, $r = -1$, and $\theta = t$, and $\phi = .5t$ when $-10 \leq t \leq 30\pi$.

6 Spherical Hypotrochoids and Epitrochoids

By modifying our spherical epicycloid/hypocycloid formula slightly, we can create an equation that plots spherical epitrochoids and spherical hypotrochoids. Epitrochoids and hypotrochoids are very similar to epicycloids and hypocycloids, but with the point that traces our path set to some fixed distance d away from the center of the rolling circle. The modified version of S_c, which we will call S_t is:

$$
\begin{aligned}
S_t = \\
&\cos(\theta)\cos(\phi)\left((R+r) - d\cos\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\right) \\
&+d\sin\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\frac{\left(\frac{\partial\theta}{\partial t}\right)\sin(\theta)\cos(\phi) + \left(\frac{\partial\phi}{\partial t}\right)\cos(\theta)\sin(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2 \cos^2(\phi)}}\,\hat{\imath} \\
&+\sin(\theta)\cos(\phi)\left((R+r) - d\cos\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\right) \\
&-d\sin\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\frac{\left(\frac{\partial\theta}{\partial t}\right)\cos(\theta)\cos(\phi) - \left(\frac{\partial\phi}{\partial t}\right)\sin(\theta)\sin(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2 \cos^2(\phi)}}\,\hat{\jmath} \\
&+\sin(\phi)\left((R+r) - d\cos\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\right) \\
&-d\sin\left(\frac{R}{r}\int_a^t \sqrt{\left(\frac{\partial\phi'}{\partial\tau}\right)^2 + \left(\frac{\partial\theta'}{\partial\tau}\right)^2 \cos^2(\phi')}\,\mathrm{d}\tau\right)\frac{\left(\frac{\partial\phi}{\partial t}\right)\cos(\phi)}{\sqrt{\left(\frac{\partial\phi}{\partial t}\right)^2 + \left(\frac{\partial\theta}{\partial t}\right)^2 \cos^2(\phi)}}\,\hat{k}
\end{aligned}
\tag{17}
$$

where $a \leq t \leq b$

The following pages contain examples of spherical epitrochoids and spherical hypotrochoids.

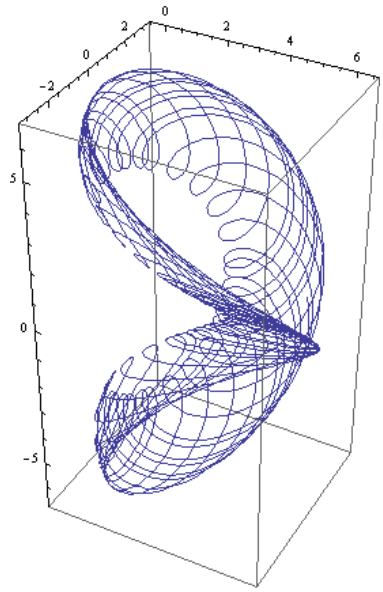


Figure 8: An example of a spherical epitrochoid. This graph was formed using our equation, S_t, with $R = 4$, $r = 1$, $\theta = \phi = t$, and $d = 1.5$ where $0 \leq t \leq 20\pi$.

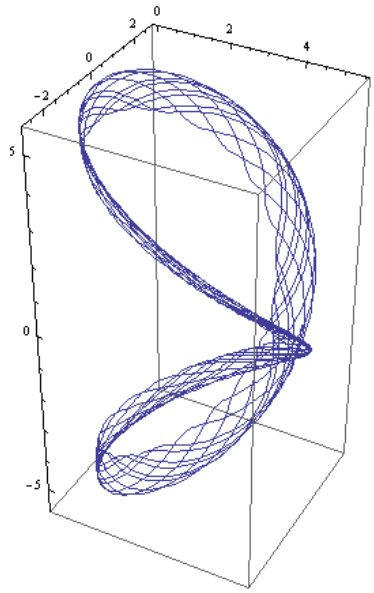


Figure 9: Another example of a spherical epitrochoid. This graph was formed using the same parameters as those used in Figure 8, but with $d = .5$.

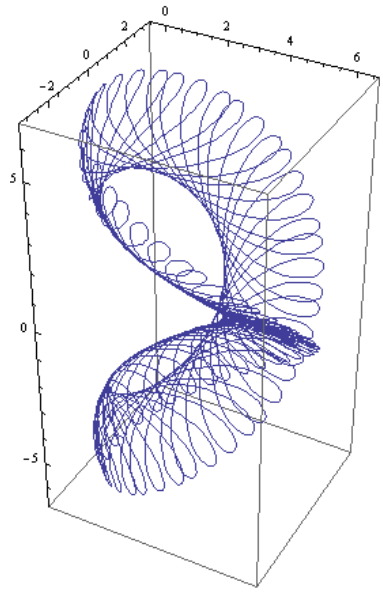


Figure 10: A spherical hypotrochoid formed using our formula when $R = 6$, $r = -1$, $\theta = \phi = t$, and $d = 1.5$ where $0 \leq t \leq 20\pi$.

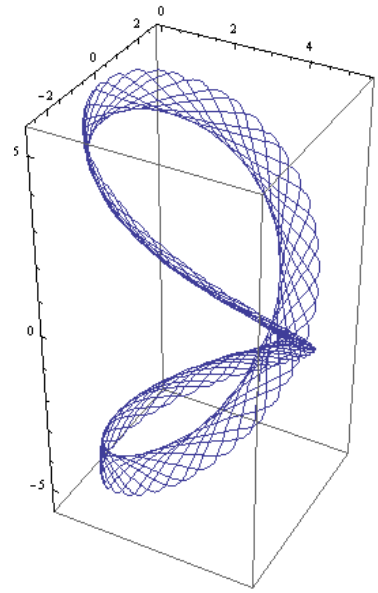


Figure 11: Another example of a spherical hypotrochoid. This graph was formed using the same parameters as those used in Figure 10, but with $d = .5$.

7 Conclusion

This paper presents formulae for spherical epicycloids, hypocycloids, epitrochoids, and hypotrochoids. These equations were born not from the need to solve some sort of engineering problem or any other problem in any given field. Instead, these equations were born from a motivation to generalize a handful of familiar equations into three dimensions simply because the idea was interesting. As such, these equations are only meant to be visually intriguing and have not been presented with a "real-world" application. Other similarly-motivated mathematicians may wish to continue this idea and generalize the epicycloid into higher dimensions.

8 Appendix A

This section demonstrates a few more curves that can be made using the formulae presented in this paper.

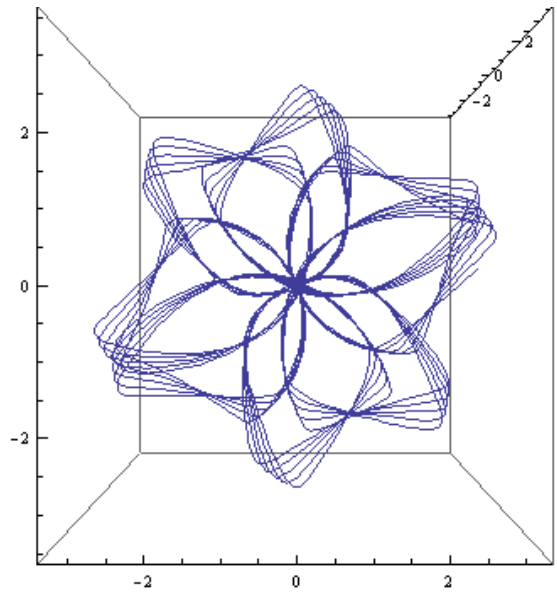


Figure 12:

Equation: S_t
$R = 4$
$r = -1$
$\theta = 3t$
$\phi = 4t$
$d = .5$
$0 \leq t \leq 10\pi$
Viewed from top.

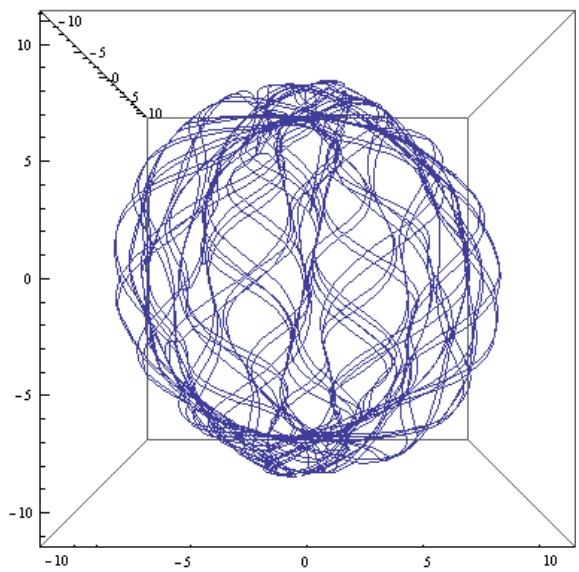


Figure 13:

Equation: S_t
$R = 4$
$r = -3$
$\theta = t$
$\phi = 4t$
$d = 10$
$0 \leq t \leq 50\pi$
Viewed from front.

Figure 14:

Equation: S_c
$R = 2$
$r = 2$
$\theta = \sin(2t)$
$\phi = \dfrac{t}{2}$
$0 \leq t \leq 50\pi$

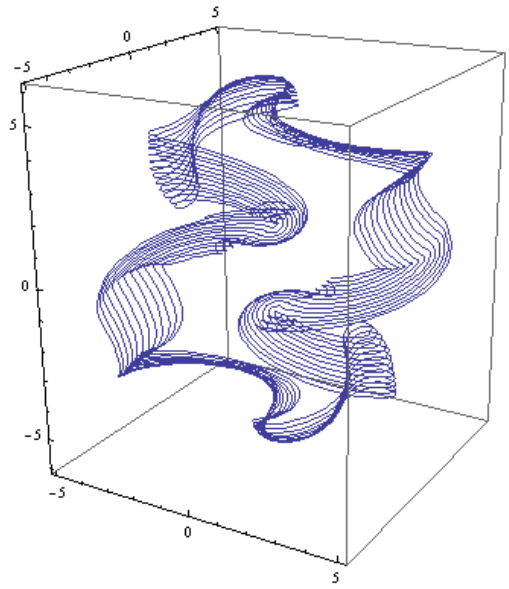
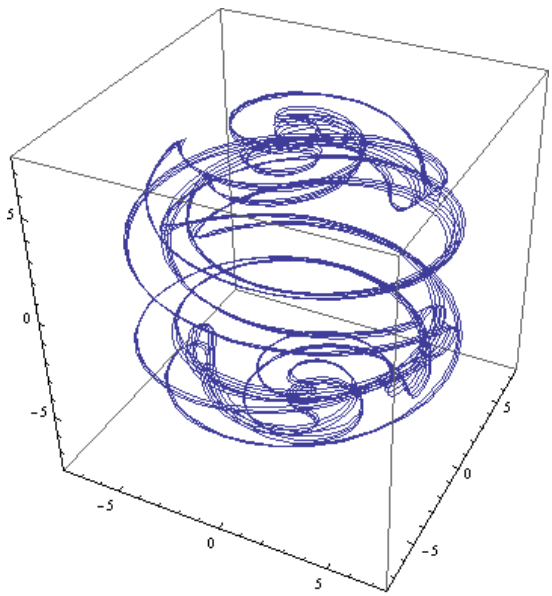


Figure 15:

Equation: S_c
$R = e$
$r = 4$
$\theta = 3\cos(2t)$
$\phi = \dfrac{t}{2}$
$0 \leq t \leq 30\pi$



9 Appendix B

All graphs seen in this paper were created using Mathematica 9.

References

[1] Weisstein, Eric W. "Cycloid." From MathWorld–A Wolfram Web Resource. http://mathworld.wolfram.com/Cycloid.html

[2] Weisstein, Eric W. "Epicycloid." From MathWorld–A Wolfram Web Resource. http://mathworld.wolfram.com/Epicycloid.html

[3] Weisstein, Eric W. "Hypocycloid." From MathWorld–A Wolfram Web Resource. http://mathworld.wolfram.com/Hypocycloid.html

[4] Weisstein, Eric W. "Epitrochoid." From MathWorld–A Wolfram Web Resource. http://mathworld.wolfram.com/Epitrochoid.html

[5] Weisstein, Eric W. "Hypotrochoid." From MathWorld–A Wolfram Web Resource. http://mathworld.wolfram.com/Hypotrochoid.html